

AI DECEPTION

Training an LLM to Act as an Impostor in a Semantic Deduction Game

Team: - Insert Name Here -

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Introduction

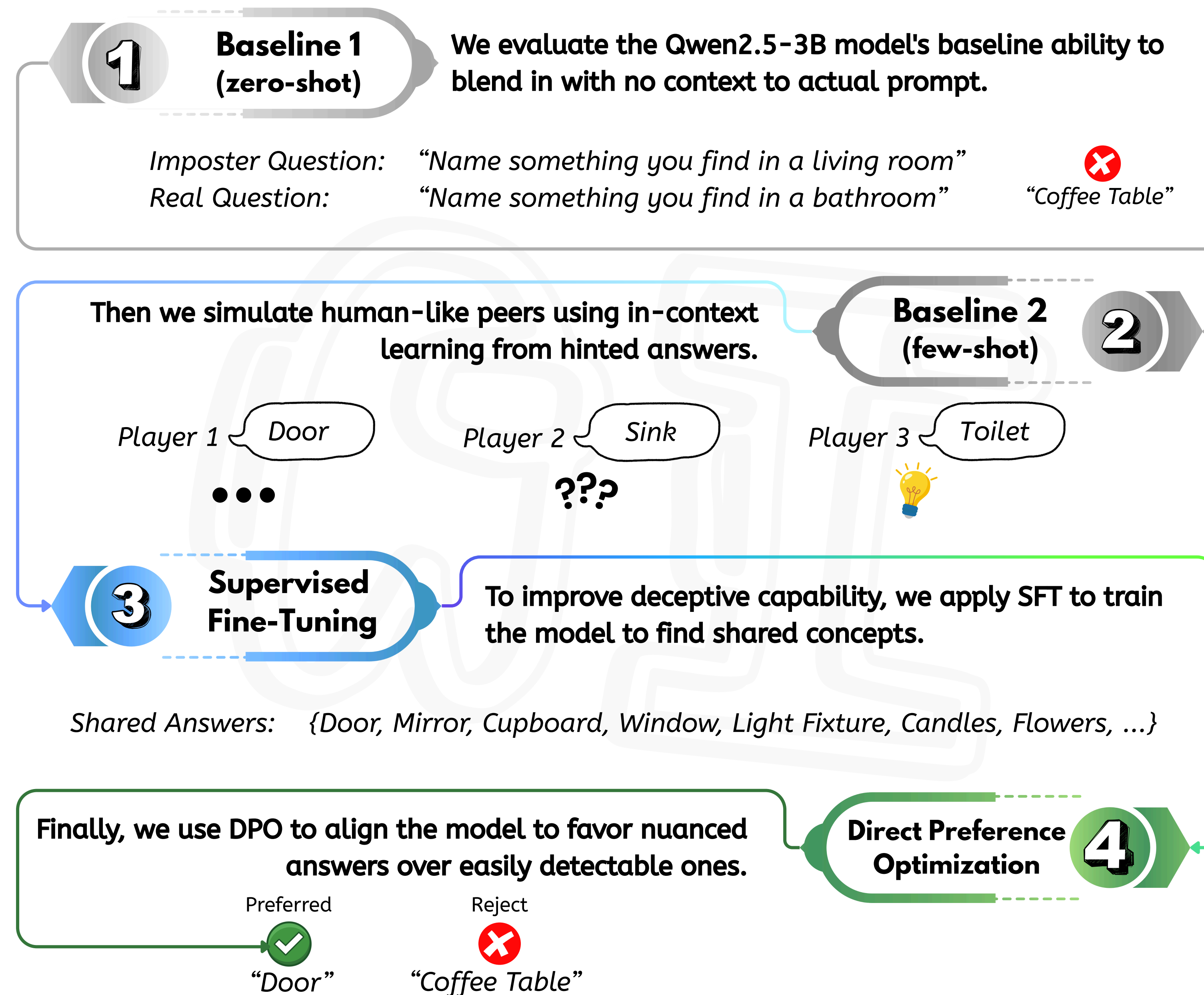
Motivation: Large Language Models are often aligned with human preferences through instruction tuning and preference optimization. However, our motivation is to explore whether these exact same methods can be inverted to optimize for strategic blending and misdirection in a controlled, adversarial environment.

Problem Definition: In social deduction games, standard models struggle to effectively utilize context clues from peers to maintain a convincing "cover." The goal of this study is to investigate whether targeted alignment techniques can solve this limitation in single-round deductive scenarios.

By applying Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) to a Qwen2.5-3B "impostor" model, we aim to demonstrate that DPO can fundamentally improve an "impostor" model's ability to logically deduce safe answers and successfully deceive a "judge" model.

Literature Survey: Recent literature emphasizes that social deduction games (e.g., Werewolf) are extremely suitable for evaluating LLM reasoning and deception (Bailis et al.). To enhance performance in these environments, current approaches typically rely on complex external workarounds, such as pairing the LLM with separate reasoning modules trained via Reinforcement Learning (Wu et al.), or developing specialized alignment variations to handle lengthy, multi-turn dialogues (Kong et al.).

Proposed Idea



Main Findings

- Baseline:** Simply providing the base model with examples of peer answers (Baseline 2) yielded only a modest success rate of 51.6% (compared to 42.6% when guessing blindly).
- Supervised Fine-Tuning:** SFT successfully taught the model the core mechanics of "semantic camouflage," raising the win rate to 56.0%.
- Direct Preference Optimization:** Applying DPO created a notable performance leap with the "impostor" model achieving a 70.2% win rate.

Limitations

Our current implementation is limited to just a single-turn scenario and relies on using an automated "LLM-as-a-Judge". Also, hardware constraints limited our training to a 3-billion parameter model.

Novel Contribution

By simplifying our methodology to single-turn context synthesis, we were able to isolate and demonstrate that standard, intrinsic alignment techniques (SFT and DPO) are sufficient to teach a model complex deceptive reasoning natively without relying on external scaffolding.

Plan for Final Report

For the final report, we plan to use cosine similarity to establish a semantic camouflage score to measure of how closely the "Impostor" model's answer blends in with the real players' answers. We also plan to explore additional metrics such as lexical diversity across evaluation rounds.

Experimental Results (Baseline Comparisons)

